

Research on Fincial Fraud Detection based on Deep Graph Neural Network

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Abstract—Financial fraud encompasses a range of dishonest practices that are designed to gain financial benefits, with the potential to adversely affect market stability, economic growth, and foster additional criminal activities. The shift of fraudulent practices to online platforms has made the detection of such activities increasingly challenging. This study introduces a graph neural network-based detection model combining Graph Convolutional Network (GCN) and Graph Attention Network (GAT), which enables the capture of intricate relationships within financial transactions. The model improves the accuracy of fraud detection by leveraging multilayer deep learning on complex patterns. The experimental results across various datasets demonstrate superior performance compared to traditional methods, suggesting strong potential for practical applications.

Keywords—Fincial fraud detection, Graph neural network, Graph attention network, Graph convolution.

I. INTRODUCTION

Financial fraud is a pervasive issue that has far-reaching consequences, affecting not only the financial sector, government, and corporate entities but also directly impacting the average consumer. The effective response and detection of financial fraud necessitates a collaborative approach between banking institutions and regulatory authorities. The financial cost of credit card fraud is estimated at billions of dollars per year, with a significant negative impact on society and business operations. The rapid development of the global economy and information technology has led to a concomitant expansion in the scale of e-commerce and online transactions^[1]. This has resulted in a parallel increase in the number and scale of financial fraud, which now represents a significant threat to social and economic stability.

The detection of fraud is frequently regarded as a matter of triage. The prevailing methods for the detection of financial fraud can be classified into two principal categories: rule-based approaches and machine learning-based approaches. Rule-based testing methods are a prevalent approach in the financial industry, relying on the expertise of financial professionals to develop detection rules. However, these methods are not without limitations^[2]. Firstly, the design of rules is dependent on the

input of financial experts, which makes it challenging for these methods to adapt to the evolving and intricate fraud patterns that are prevalent in the financial sector. Furthermore, rule-based methods necessitate a considerable investment of human resources for analysis and evaluation. For those without specialist expertise, the process of identifying fraud can be lengthy and require substantial training, which can lead to inefficiencies in detection^[3].

Financial fraud represents a significant global issue, resulting in substantial financial losses and a crisis of confidence between financial institutions, businesses, and consumers. The accelerated growth of financial markets and the advent of sophisticated technology have led to the emergence of increasingly complex and challenging fraud techniques. Conventional financial fraud detection methodologies rely predominantly on expert knowledge^[4]. Despite their long-standing usage, these methods exhibit inherent limitations in terms of design flexibility and their capacity to adapt effectively to the rapid evolution of fraud techniques. Furthermore, rules-based approaches necessitate substantial human involvement, which impedes the capacity to respond effectively to novel or unidentified frauds.

In order to address these challenges, machine learning-based methods are being introduced into the field of financial fraud detection. The majority of machine learning methods perform predictive classification by extracting a multitude of statistical features from users, and these methods are particularly adept at automatically learning potential fraud patterns, thereby markedly enhancing the efficiency and accuracy of detection. However, a significant number of machine learning methods concentrate on the statistical attributes of a single user, thereby neglecting the interconnectivity between users^[5]. This has the consequence of imposing certain constraints on the capacity of these methods to effectively identify complex fraud. This limitation is particularly evident in the context of complex financial transaction networks.

To overcome these limitations, machine learning-based methods are introduced into financial fraud detection. Most of these methods extract the statistical characteristics of users from different aspects and use these features for predictive

classification [6]. However, these methods rarely take into account the interaction between users. Machine learning methods greatly improve the efficiency and accuracy of detection by analyzing historical data to automatically learn and identify potential fraud patterns. However, most machine learning methods mainly focus on the statistical characteristics of individual users and ignore the interaction between users, which limits their detection ability to a certain extent [7].

The complexity and volatility of financial markets make graphs built on financial data often heterogeneous or time-changing, which poses challenges to modeling techniques. Financial data is complex and diverse, and is usually presented in the form of graph data, but traditional financial fraud detection methods cannot effectively process this kind of graph data [8]. The advent of graph neural networks (GNNs) solves this problem well. Graph neural networks are a deep learning method that operates on graph domains and are often designed for tasks.

II. RELATED WORK

Over the past few decades, the operating model based on embedded rules has become very popular in the financial community. Because these rules are simple and easy to code, they are often developed by consultants and domain experts who apply their own work experience to automated decision-making processes. Roldán-García et al. [9] proposed a rule-based expert system whose main purpose is to check and manage anti-fraud rule datasets to avoid semantic conflicts that lead to erroneous execution of the underlying expert system. An expert system is defined as a computing system that is capable of representation and reasoning within a specific domain of knowledge in order to solve problems and provide recommendations.

In the field of fraud detection, one of the traditional machine learning methods is Support Vector Machine (SVM) [10], which has been widely used in the field of fraud detection and anti-money laundering. Although SVMs can achieve good detection results, they take a long time to train. To address data imbalances and long training times, SVMs can be optimized using hyperparameter tuning and random sampling, a combination of which can compensate for the shortcomings of a single approach. Bayesian classifiers [11] are also one of the machine learning methods commonly used for fraud detection. In fraud detection, logical operators and if-else conditions can be used to analyze the relationships between attributes in a dataset, while Bayesian classifiers can effectively identify and find money laundering activities.

Le et al. [12] proposed a method that combines clustering with multilayer perceptron (MP) to detect suspected money laundering cases. The technique uses a simple center-based clustering technique that uses the frequency and number of transactions as input to the multilayer perceptron training process. The principle of the unsupervised detection algorithm is to predict outliers by searching for outliers. However, these algorithms are often criticized for their high false-positive and false-negative rates in real-world deployments. To solve this problem, Shubhomoy et al. [13] proposed an active anomaly discovery method combined with expert feedback, by adjusting the anomaly detector to make the outliers it finds more in line with the experts' semantic understanding of the anomaly.

In recent years, cryptocurrencies have become a haven for money laundering activities, however, existing unsupervised anomaly detection methods are insufficient in detecting real Bitcoin transactions. To solve this problem, Joana et al. [14] proposed an active learning solution based on a real Bitcoin transaction dataset that can match the performance of a supervised baseline algorithm with only 5% labels.

III. METHODOLOGIES

A. Notions

Initially, we summarize the primary used parameters and its functions in following Table 1.

TABLE I. PRIMARY NOTIONS

Symbols	Functions
$G = (V, E)$	Preprocessed graph data
$\tilde{A}$	Adjacency matrix plus the self-loop
$\tilde{D}$	Degree matrix
$H^{(l)}$	Node feature matrix
$h_j^{(l)}$	Eigenvector
$a^{(l)}$	Attention weight vector
$[\cdot \cdot]$	Splicing operation of the vector
y	Prediction result

B. Graph embedding

The features are extracted from the financial transaction records and the data structures of nodes and edges are constructed. Nodes represent transaction entities, and edges represent relationships between entities. Each node and edge is accompanied by an eigenvector that represents its properties and relationship information. Preprocessed data is constructed into a graph $G = (V, E)$, where V represents node set and E represents the edge set. Each node $v_i \in V$ corresponds to an eigenvector x_i , and each edge $e_{ij} \in E$ corresponds to an eigenvector e_{ij} .

GCN aggregates information about each node and its neighbors to build associations between nodes to identify complex fraud patterns in the transaction network. This aggregation method allows the model to analyze transaction behavior from a global perspective, focusing not only on individual nodes, but also on the interaction between nodes. At the same time, GAT has introduced an attention mechanism that assigns different weights to nodes based on their importance, especially assigning higher weights to high-risk nodes.

C. Graph convolution and attention networks

The basic idea of a graph convolutional neural network is to aggregate information from neighbor nodes by performing convolution operations through adjacency matrices. Specifically, the hierarchical update of the graph convolutional neural network is shown in Equation 1:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right) \quad (1)$$

where $\tilde{A} = A + I$ denotes the adjacency matrix plus the self-loop, $\tilde{D}$ is the degree matrix of $\tilde{A}$, $H^{(l)}$ is the node feature matrix

of the l layer, $W^{(l)}$ is the weight matrix of the l layer, and σ is the activation function.

Additionally, the graph attention network introduces an attention mechanism that aggregates information by assigning different weights to different neighbor nodes. The updated calculation is expressed as following Equation 2.

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij}^{(l)} W^{(l)} h_j^{(l)} \right), \quad (2)$$

where $W^{(l)}$ represents the weight matrix for layer l . The eigenvector is expressed as $h_j^{(l)}$ node j at the l -layer. The $\alpha_{ij}^{(l)}$ is the attention coefficient, which is calculated by the Equation 3.

$$\alpha_{ij}^{(l)} = \frac{\exp(\text{LeakyReLU}(a^{(l)T} [W^{(l)} h_i^{(l)} \| W^{(l)} h_j^{(l)}])))}{\sum_{k \in N(i)} \exp(\text{LeakyReLU}(a^{(l)T} [W^{(l)} h_i^{(l)} \| W^{(l)} h_k^{(l)}]))}, \quad (3)$$

where $a^{(l)}$ represents the attention weight vector of layer l . $a^{(l)T}$ transpose of the attention weight vector at the level l . The eigenvector of the $W^{(l)} h_i^{(l)}$ node i after being transformed by the weight matrix at level l . The eigenvector of the $W^{(l)} h_j^{(l)}$ node j after being transformed by the weight matrix at the l layer. Operation function $[\cdot \| \cdot]$ represents the splicing operation of the vector. *LeakyReLU* activation function, Leaky ReLU is a variant of ReLU that is used to calculate attention weights. $\sum_{k \in N(i)} (\cdot)$ normalize all neighbor k of node i .

In above method, the graph attention network can effectively aggregate the neighbor information of nodes and assign different weights according to the similarity between nodes, so as to better capture the complex graph structure information. Finally, we input the output eigenvectors of the graph neural network into a fully connected layer and classify them through the Softmax function to detect fraud, which is expressed as following Equation 4.

$$y = \text{Softmax}(W_o h_i + b_o), \quad (2)$$

where W_o and b_o are the weights and biases of the output layer, and y is the prediction result. Mini-batch and distributed training methods are used to maintain high processing efficiency in the constantly updated data stream.

IV. EXPERIMENTS

A. Experimental setups

In this experiment, open-source datasets from Yelp and Amazon were used to verify the effectiveness of the proposed ER-GNN model in fraud detection. The Yelp dataset focuses on spam comment detection and contains three main relationships: reviews posted by the same user (R-U-R), reviews under the same product with the same star rating (R-T-R), and reviews published by the same product in the same month (R-S-R). Amazon datasets also have a similar ternary relationship structure that helps identify potential fraud in reviews. These datasets are represented in the form of graphs, with nodes representing comments and edges representing relationships between comments.

Due to the large number of dataset nodes, the mini-batch training method was used to train ER-GNN and other baseline algorithms. AT-GCN uses the Adam optimizer and the learning rate is set to 0.01. In ER-GNN, the number of training rounds is 50, the number of layers is 1, the regularization parameter is 2, and the action step size for reinforcement learning is 0.02. In the Yelp dataset, the model has a batch size of 1024 and a learning rate of 0.01, and in the Amazon dataset, the batch size is 256 and a learning rate of 0.005. These settings ensure training efficiency and model performance when working with large-scale data.

B. Experimental analysis

In financial fraud detection, Accuracy is an important evaluation metric, which measures the model's ability to correctly classify all samples, including fraud and non-fraud samples. Accuracy is defined as the number of samples correctly classified as a proportion of the total sample size. Following Figure 1 demonstrates detection accuracy with existing methods.

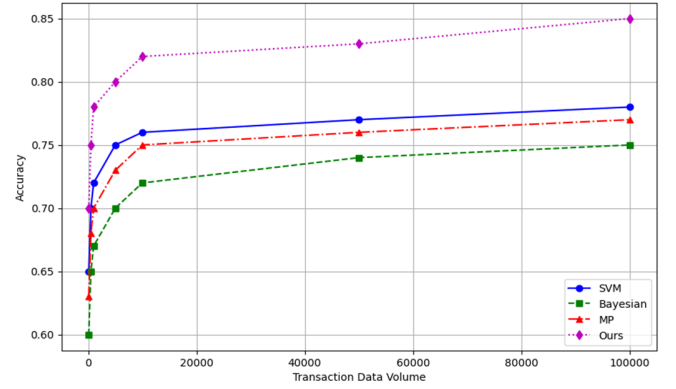


Figure 1. Accuracy Comparison of Different Methods.

In financial fraud detection, the impact of different parameter settings on model performance is crucial. We conducted parameter sensitivity experiments on the dataset, focusing on the influence of different loss function weights on the performance of the model. To verify the robustness of our method, we conducted parameter sensitivity experiments on the Elliptic dataset to investigate the impact of different loss function weights on model performance. In the experiment, the weight of the normal node gradually increases from 0.1 to 0.9.

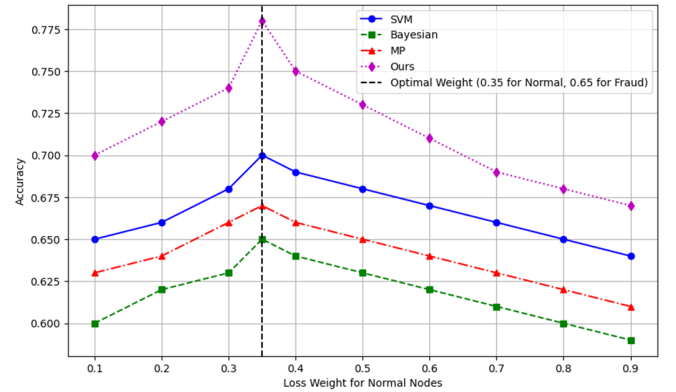


Figure 2. Model Performance with Different Loss Weights.

Above Figure 2 shows the impact of different loss function weights on the performance of each method model on the dataset. The abscissa is the weight assigned to the normal node, and the ordinate is the accuracy of the model. Methods of comparison include SVM, Bayesian, MP, and our method. As you can see, our method achieves the highest accuracy when the weight of the normal node is 0.35. Through this parameter sensitivity experiment, the robustness of each model under different parameters can be verified, and the optimal parameter configuration can be found to improve the model performance.

Recall is the proportion of actual fraud that the model successfully detects. It reflects the model's ability to catch fraud. High recall rates mean that the model is able to detect the majority of fraud, reducing the risk of undetected fraud, reducing potential financial losses. Following Figure 3 compares the average recall comparison results among different model.

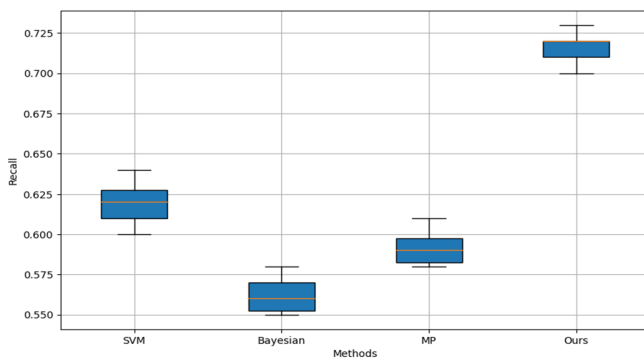


Figure 3. Recall Comparison of Different Methods

The above Figure 3 shows an experimental comparison of different methods on recall. The abscissa is for different methods, including SVM, Bayesian, MP, and ours, and the ordinate is the recall. Table 2 compares the precision and recall of different fraud detection methods. The results show that the GNN model is significantly better than the traditional method in terms of precision and recall.

TABLE II. BASIC SITUATION OF DOMESTIC CPU TECHNOLOGY

Dataset	Method	Precision (%)	Recall (%)
Yelp	SVM	78%	65%
	Bayesian Classifier	72%	60%
	Center Clustering & MP	80%	67%
	GNN Model	89%	79%
Amazon	SVM	75%	63%
	Bayesian Classifier	70%	58%
	Center Clustering & MP	77%	66%
	GNN Model	88%	76%

V. CONCLUSIONS

In conclusion, a financial fraud detection method based on Deep Graph Neural Network is proposed, and its effectiveness is verified by experiments. Through experimental comparisons on multiple open-source datasets, it was found that our method significantly outperformed traditional methods in terms of accuracy and recall. The parameter sensitivity experiment further proves that the appropriate parameter setting can significantly improve the performance of the model, and verifies the robustness and practicability of the model. Future work will further optimize the model structure and explore more graph neural network variants to improve detection performance and apply it to more real-world scenarios.

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